

If you want to repeat the sequence more than once, you can also use the setting **Material selection > Sequence Cyclical**.

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Simulating the Accrued Costs of the Machines

Simulate the Accrued Costs of the Machines

You can simulate the accrued costs of the machines, which you entered on the tab **Costs**, with the object *CostAnalyzer*.

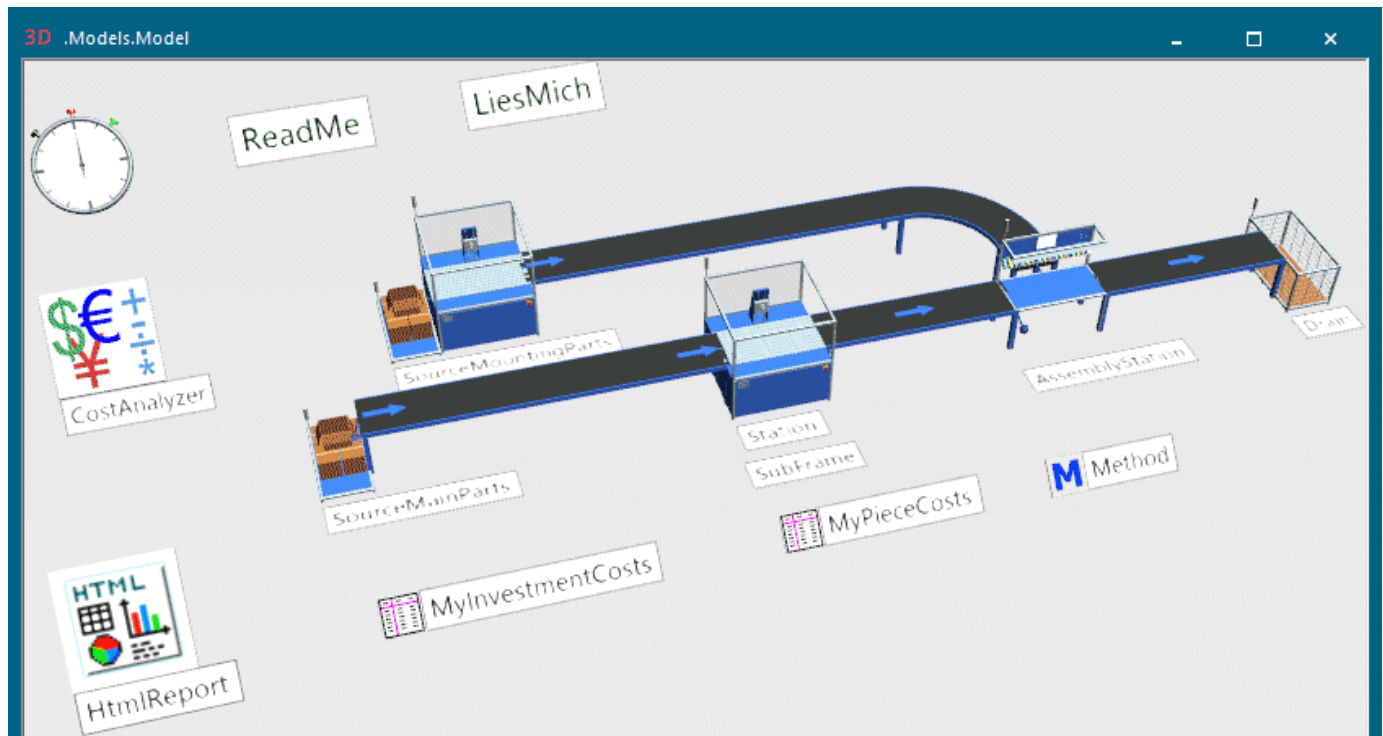
With the object **CostAnalyzer**  you can analyze the costs that accrue for the *active material flow objects*, for the **Part** and for the **Container** in your plant.

You can enter the respective costs on the tab **Costs** of the *material flow objects*, and on the tab **Costs** of the *Part* and of the *Container*.

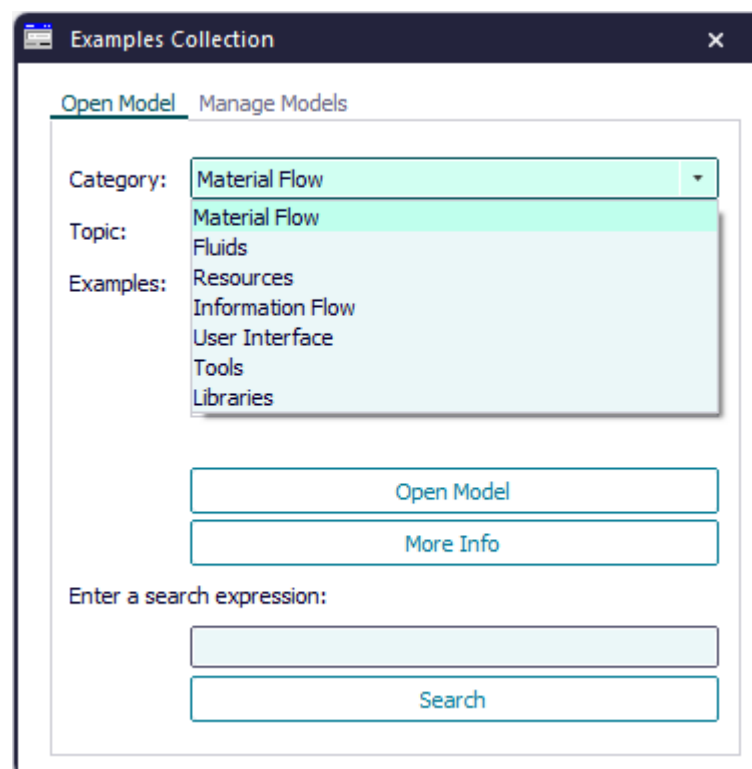
We will demonstrate how to:

- [Specify the Costs of the Conveyors](#)
- [Specify the Costs of the Stations and the Parts](#)
- [Configure the Costs Report](#)
- [View the Costs Report](#)

The finished simulation model looks like this:



Compare the sample models: Click the **Window** ribbon tab, click **Start Page > Getting Started > Example Models > Small Examples**. Then, select the respective **Category**, the **Topic**, and the **Example** in the dialog **Examples Collection**, and click **Open Model**.

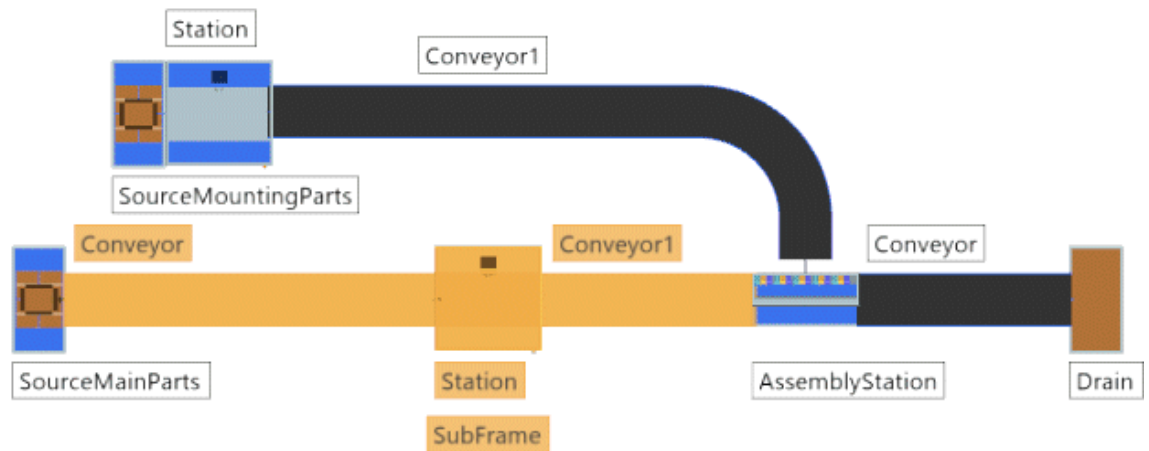


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Specify the Costs of the Conveyors

We also enter the costs which the *Conveyors* create. To show how the costs from *sub-Frames* are collected and displayed, we modeled part of our model in the *sub-Frame* named SubFrame.

While creating our model, we also enter the costs which the *Conveyors* create. To show how the costs from *sub-Frames* are collected and displayed, we modeled part of our model in the *sub-Frame* named SubFrame. To show the content of the *sub-Frame*, we activated **Show Content** on the tab **Graphics** of the *sub-Frame*.



For *Conveyor1* we typed in these costs:

The screenshot shows the **.Models.Model.Conveyor1** dialog box with the **Costs** tab selected. The **Name** field is set to **Conveyor1**. The **Label** field is empty. The **Planned** checkbox is checked. The **Entrance Locked** and **Exit Locked** checkboxes are unchecked. The **Active** checkbox is checked. The **Depreciation Period** is set to 10 years. The **Investment Costs** are set to 15000 €. The **Investment Costs per Length** are set to 0 €/m. The **Operating Costs** are set to 2000 €/year. The **Operating Costs per Length** are set to 0 €/year × m.

Parameter	Value	Unit
Depreciation Period	10	years
Investment Costs	15000	€
Investment Costs per Length	0	€/m
Operating Costs	2000	€/year
Operating Costs per Length	0	€/year × m

For the *Conveyor* in the *sub-Frame* we typed in these values:

The screenshot shows the **.Models.CostAnalyzer.GettingStarted.SubFrame.Conveyor** dialog box with the **Costs** tab selected. The **Name** field is set to **Conveyor**. The **Label** field is empty. The **Planned** checkbox is checked. The **Entrance Locked** and **Exit Locked** checkboxes are unchecked. The **Active** checkbox is checked. The **Depreciation Period** is set to 8 years. The **Investment Costs** are set to 2000 €. The **Investment Costs per Length** are set to 0 €/m. The **Operating Costs** are set to 200 €/year. The **Operating Costs per Length** are set to 0 €/year × m.

Parameter	Value	Unit
Depreciation Period	8	years
Investment Costs	2000	€
Investment Costs per Length	0	€/m
Operating Costs	200	€/year
Operating Costs per Length	0	€/year × m

For *Conveyor1* in the *sub-Frame* we typed in these values:

The screenshot shows the 'Costs' tab of the 'Conveyor1' sub-frame. The 'Active' checkbox is checked. The following cost parameters are set:

Parameter	Value	Unit
Depreciation Period	10	years
Investment Costs	2000	€
Investment Costs per Length	0	€/m
Operating Costs	200	€/year
Operating Costs per Length	0	€/(year × m)

For the *Conveyor* connecting the *AssemblyStation* with the *Drain* we typed in these values:

The screenshot shows the 'Costs' tab of the 'Conveyor' dialog box. The 'Active' checkbox is checked. The following cost parameters are set:

Parameter	Value	Unit
Depreciation Period	8	years
Investment Costs	1500	€
Investment Costs per Length	0	€/m
Operating Costs	150	€/year
Operating Costs per Length	0	€/(year × m)

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Specify the Costs of the Stations and the Parts

We also enter the costs that the *Stations* create. To show how the costs from *sub-Frames* are collected and displayed, we modeled part of our model in the *sub-Frame* named *SubFrame*.

To show the content of the *sub-Frame* instead of its icon, we activated **Show Content** on the tab **Graphics** of the *sub-Frame* in 3D.

- For *Station1* we typed in these values:

The screenshot shows the **.Models.CostAnalyzer.GettingStarted.Station** dialog box. The **Costs** tab is active. The **Active** checkbox is checked. The **Depreciation Period** is set to 1 years, **Investment Costs** to 0 €, and **Operating Costs** to 2000 €/year. Other tabs include Times, Set-Up, Failures, Controls, Exit, Statistics, Importer, and Energy.

In addition we typed in a **constant processing time** of 10 minutes:

The screenshot shows the **.Models.Model.Station** dialog box. The **Cost** tab is active. The **Processing Time** is set to **Const** with a value of 10:00. The **Automatic Processing** checkbox is checked. Other tabs include Times, Set-Up, Failures, Controls, Exit, Statistics, and Importer.

- For the *Station* in the *sub-Frame* we typed in these values:

The screenshot shows the 'Costs' tab of the 'Station' configuration window. The 'Name' field is 'Station' and 'Label' is empty. The 'Planned' status is selected. The 'Costs' tab is active, showing the following values:

Parameter	Value	Unit
Depreciation Period	10	years
Investment Costs	4500	€
Operating Costs	200	€/year

In addition we typed in a **constant processing time** of 15 minutes:

The screenshot shows the 'Times' tab of the 'Station' configuration window. The 'Processing Time' is set to 'Const' with a value of '15:00'. The 'Automatic Processing' checkbox is checked.

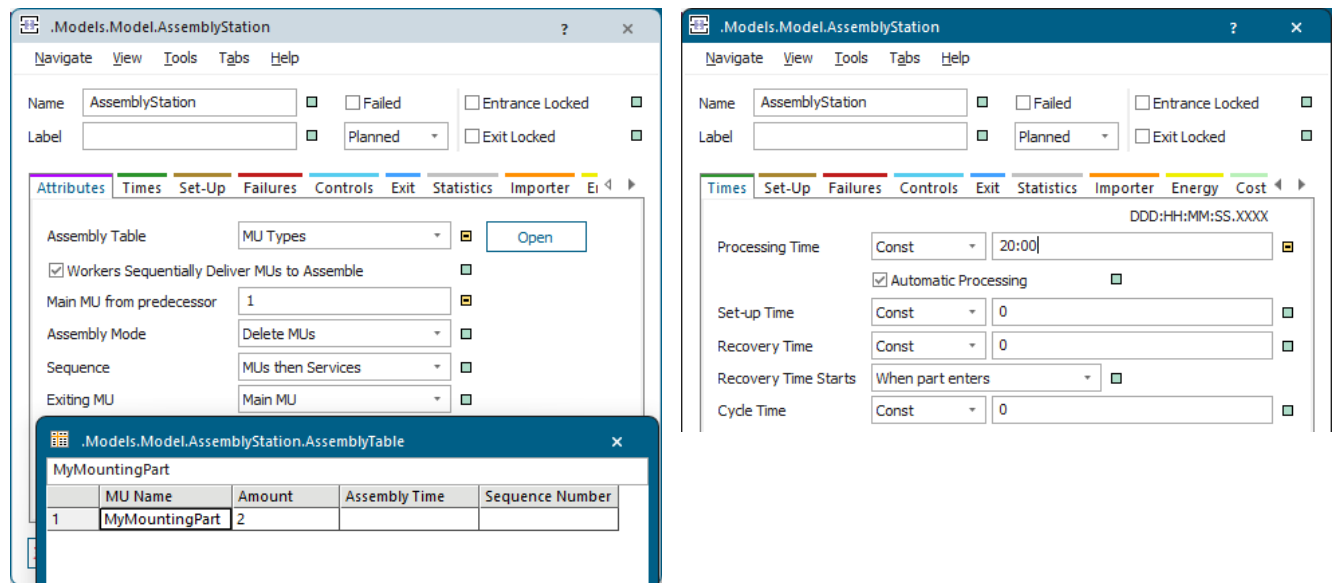
Parameter	Value
Processing Time	Const 15:00

- For the *AssemblyStation* we typed in these values:

The screenshot shows the 'Costs' tab of the 'AssemblyStation' configuration window. The 'Name' field is 'AssemblyStation' and 'Label' is empty. The 'Planned' status is selected. The 'Costs' tab is active, showing the following values:

Parameter	Value	Unit
Depreciation Period	1	years
Investment Costs	0	€
Operating Costs	1000	€/year

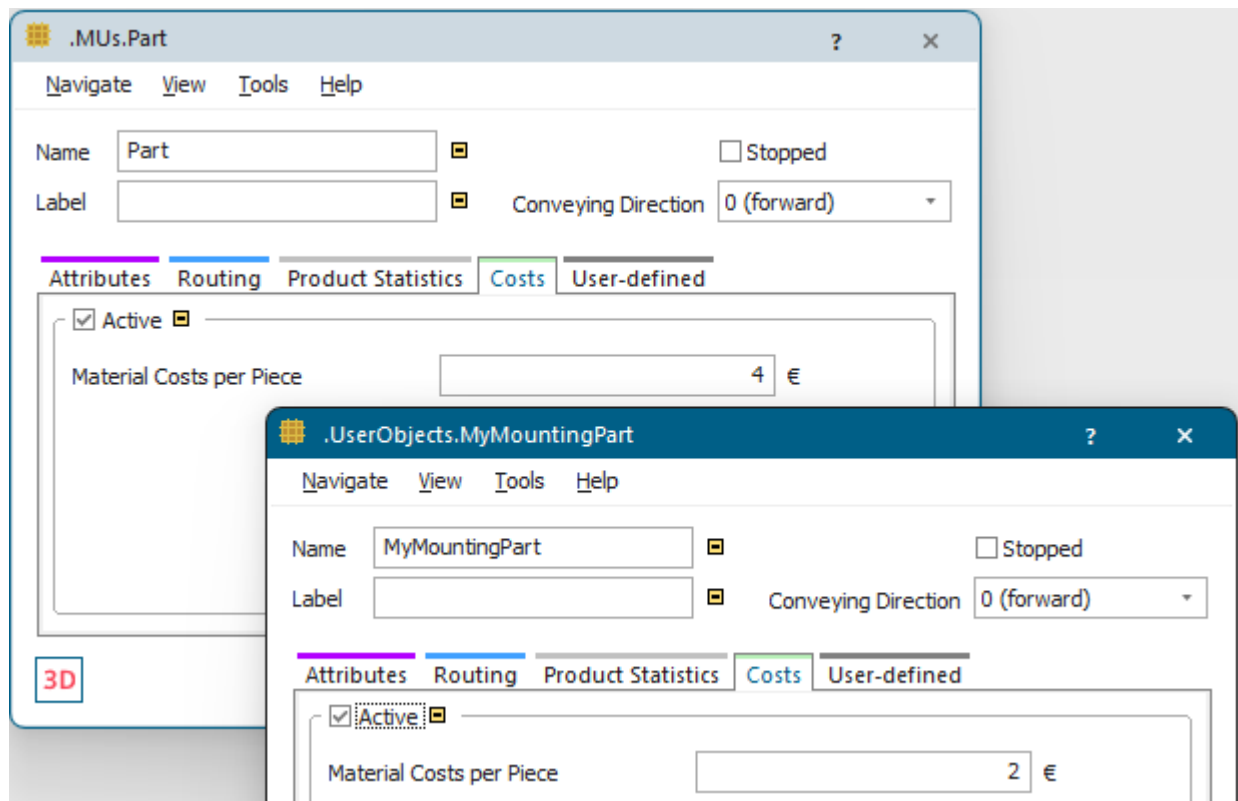
We also selected these settings:



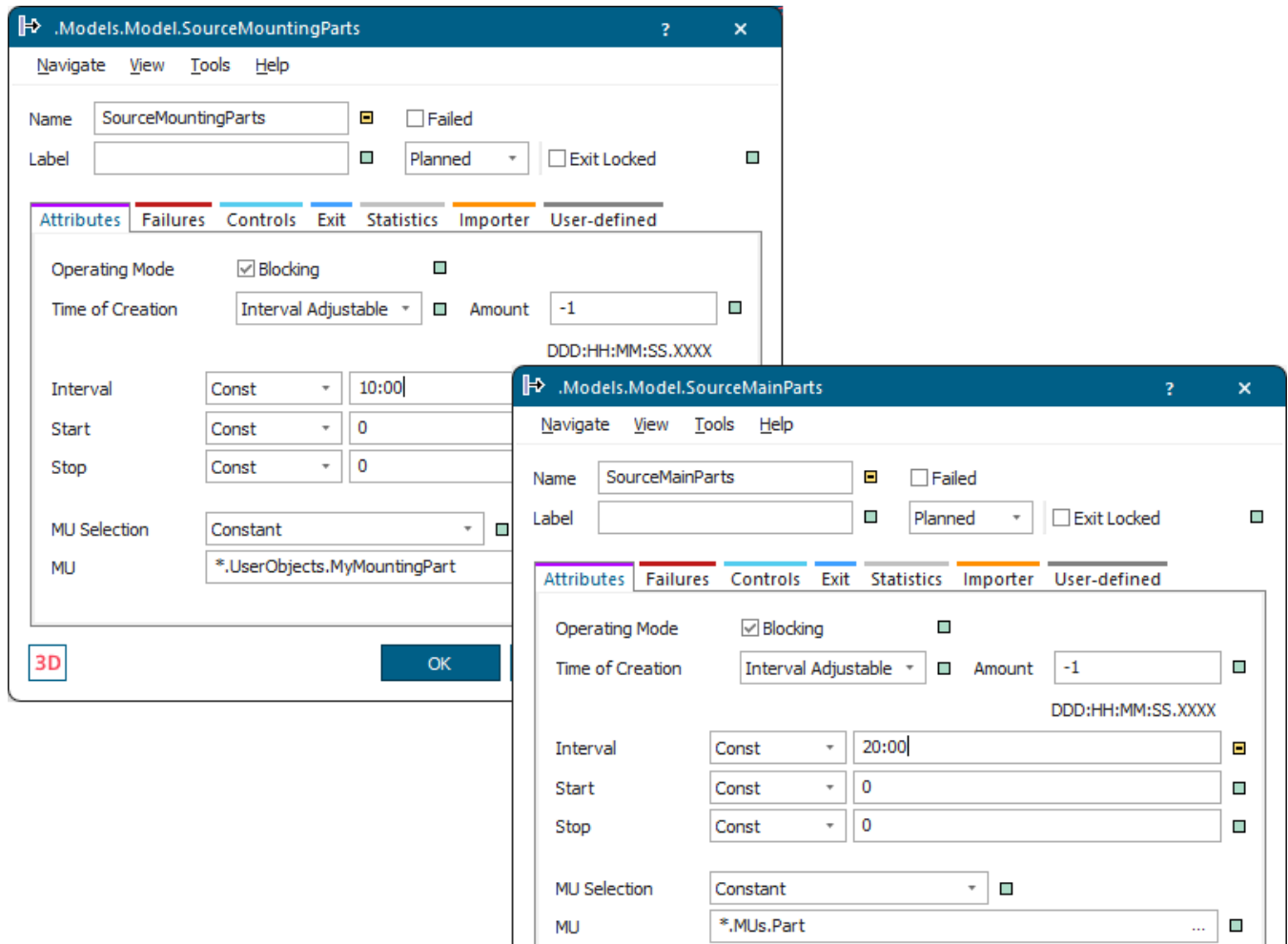
Note

Only type the **name** of the mounting part into the **Assembly Table**, not the entire path.

- Next we entered the costs for our part types, namely for the *Part* and for *MyMountingPart*:



- Finally we configured our Sources named SourceMainParts and SourceMountingParts by typing in a creation **interval** each:

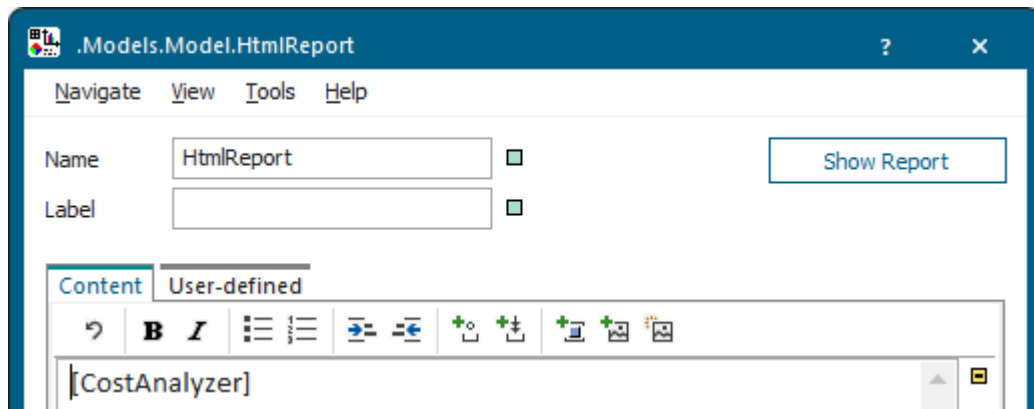


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Configure the Costs Report

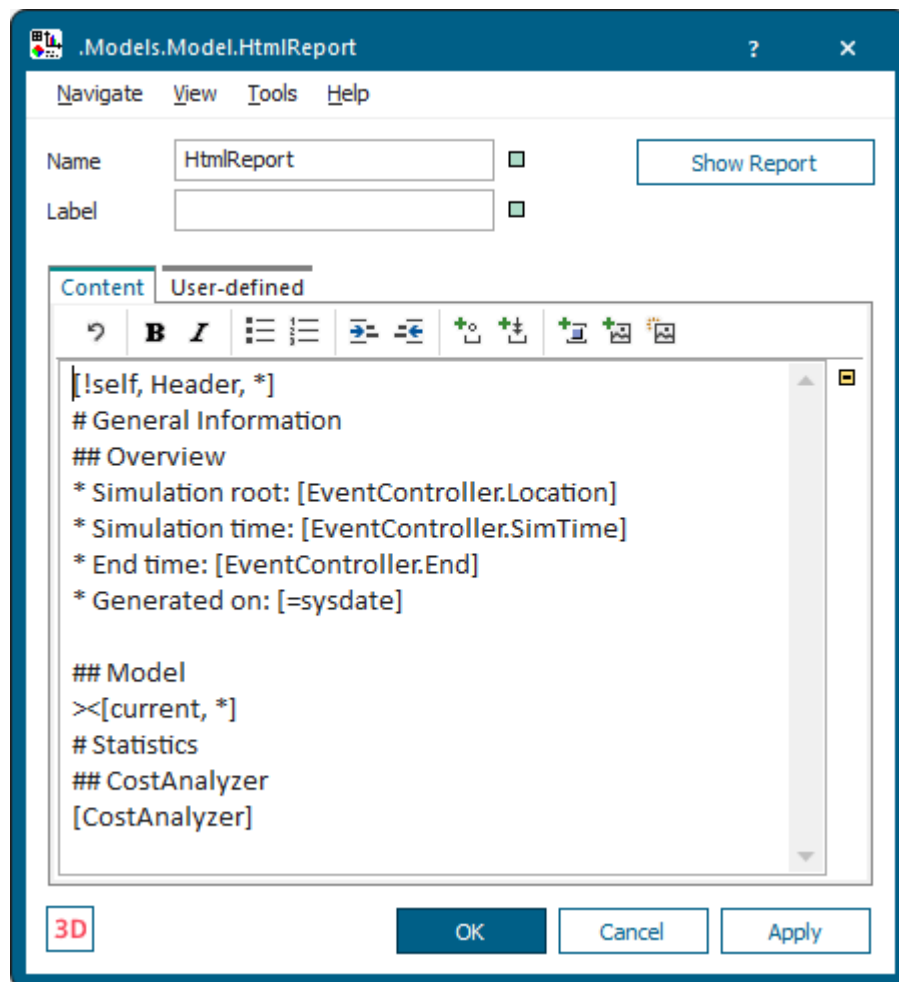
We cannot configure the *CostAnalyzer* as such. To show the results of the cost analysis, **drag** the *CostAnalyzer* from the *Frame* onto the icon of the *HtmlReport* **and drop** it there.

- If the **Content** of the [HtmlReport](#) is inherited, the data of the *CostAnalyzer*  replaces the pre-defined content of the *HtmlReport*. *Plant Simulation* opens the display window of the *HtmlReport* automatically.



- If the **Content** of the *HtmlReport* is not inherited, *Plant Simulation* appends the data of the *CostAnalyzer* to the pre-defined content of the *HtmlReport*. *Plant Simulation* opens the display window of the *HtmlReport* automatically.

To show the results of the cost analysis, you can also drag the object *CostAnalyzer* into the open window of the *HtmlReport* and drop it there, for example below the heading **Statistics**. Then the *HtmlReport* also shows the pre-defined **General Information** when you click **Show Report**.



This adds the following instruction to the *HtmlReport*:

```
# Statistics
## CostAnalyzer
[CostAnalyzer]
```

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View the Costs Report

The *HtmlReport* shows the costs report with the settings we defined.

The screenshot shows the **Statistics** section of the **CostAnalyzer** report. It contains two tables: **Investment costs** and **Piece costs**.

Investment costs table:

Object	Investment costs [€]	Depreciation period [years]	Operating costs [€/year]
Model	25000.00		5750.00
└ SubFrame	8500.00		600.00
└ Conveyor	2000.00	8.00	200.00
└ Station	4500.00	10.00	200.00
└ Conveyor1	2000.00	10.00	200.00
└ AssemblyStation	0.00	1.00	1000.00
└ Conveyor	1500.00	8.00	150.00
└ Station	0.00	1.00	2000.00
└ Conveyor1	15000.00	10.00	2000.00

Piece costs table:

MU Type	Piece costs [€]	Material costs [€]	Accrued costs [€]	General costs [€]	Throughput	Work in process
Part	8.32	8.00	0.17	0.15	10650	6

- In the table **Investment Costs** the *Report* shows the summed up investment costs of this *Frame* in the unavailable cells.

Below it lists the **investment costs**, the **depreciation period**, and the **operating costs** of the individual objects inserted into this *Frame*, row by row.

You can also query the **investment costs** with the method **putInvestmentCostsIntoTable**.

- In the table **Piece Costs** the *Report* shows the summed up piece costs of the used **MU type**, in our case of the *Part*.
 - The **MU Type** (part type) designates the name of the parts when they arrive in the *Drain*. When parts change their name during the simulation, for example during the different production phases, the costs are assigned to the final product.

- The **Piece costs** proper is the sum of the **material costs**, the **accrued costs**, and the **general costs** are 8.32 €.
- The **Material costs** designate the costs, which accrue during the simulation when a new part enters the production, are 8.00 €.
- The **Accrued costs** designate the costs, which are assigned to the part while it is processed on the material flow object, are 0.17 €.
- The **General costs** designate the costs, which accrue when a material object is empty during the simulation, are 0.15 €. These costs are distributed across the part types which were processed by the material flow object.
- The **Throughput** designates the number of parts which were processed and moved through the production line, 10650 parts in our case.
- The **Work in process** designates the number of parts which are currently passing through the production process, 6 parts in our case.

Click **Investment Costs** in the dialog of the *CostAnalyzer* to show the investment costs.

.Models.Model.CostAnalyzer.getInvestmentCostsTable				
	Object	Investment costs	Depreciation period	Operating costs
1	*.Models.Model	25000.00		5750.00
2	*.Models.Model.SubFrame	8500.00		600.00
3	*.Models.Model.SubFrame.Conveyor	2000.00	8.00	200.00
4	*.Models.Model.SubFrame.Station	4500.00	10.00	200.00
5	*.Models.Model.SubFrame.Conveyor1	2000.00	10.00	200.00
6	*.Models.Model.AssemblyStation	0.00	1.00	1000.00
7	*.Models.Model.Conveyor	1500.00	8.00	150.00
8	*.Models.Model.Station	0.00	1.00	2000.00
9	*.Models.Model.Conveyor1	15000.00	10.00	2000.00

You can also query the investment costs with the method `putInvestmentCostsIntoTable`.

Click **Piece Costs** in the dialog of the *CostAnalyzer* to show the piece costs.

.Models.Model.CostAnalyzer.getPieceCostsTable							
	MU Type	Piece costs	Material costs	Accrued costs	General costs	Throughput	Work in process
1	Part	8.32	8.00	0.17	0.15	10650	6

You can also query the piece costs with the method `putPieceCostsIntoTable`.

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Simulating the Power Consumption in Your Facility

Simulate the Power Consumption in Your Facility

To accommodate the growing environmental awareness, *Plant Simulation* provides features to keep track of energy consumption in the face of rising energy costs and the depletion of energy resources.

The features on the **Tab Energy** provide these evaluations regarding energy consumption during the production process in the plant:

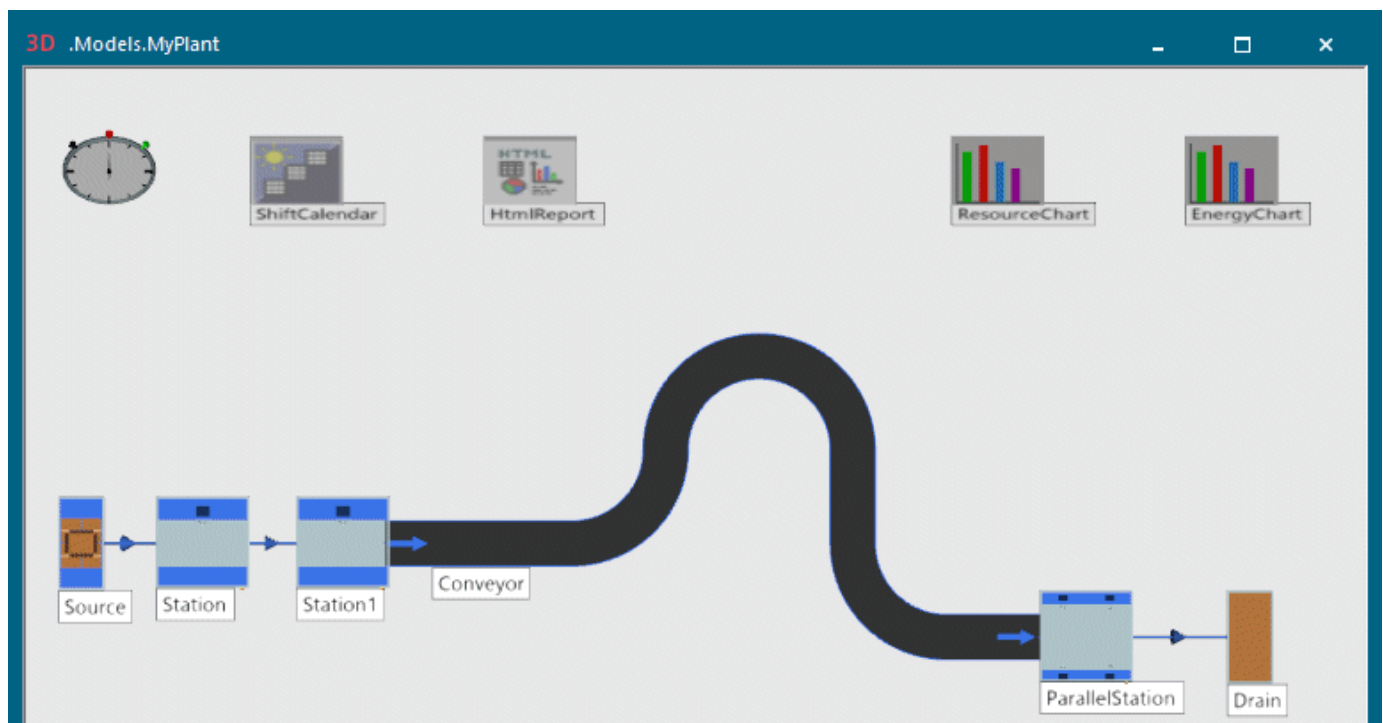
- Power consumption per machine
- Total power consumption for all objects contained within a *Frame*

In our example we will demonstrate how to:

- **Configure the Processing Stations and the Conveyor**
- **Configure the ShiftCalendar**
- **Check the Power Consumption in the Dialogs of the Objects**
- **Check the Power Consumption in the Statistics Report**
- **Check the Power Consumption in the Chart**

We use most of the material flow objects with their default settings. We just entered energy-related settings and defined failures for some of the objects.

The finished simulation model looks like this:



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Configure the Processing Stations and the Conveyor

Insert the following stations to simulate our parts moving through the factory.

- The *Source* feeds the parts into the plant. We did not change any of its default settings.
- The first machine, the *Station*, processes the parts for 1 minute each. We entered the kilowatts it uses up in the states it can be in. We also entered the times that the machine requires for switching from one energy state to another.

.Models.MyPlant.Station

Navigate View Tools Tabs Help

Name: Station ☐ Failed ☐ Entrance Locked ☐

Label: ☐ Planned ☐ Exit Locked ☐

Times Set-Up Failures Controls Exit Statistics Importer **Energy** Cost

☒ Active

Power Input [kW]	
Working	1
Setting-up	1
Operational	1
Failed	1
Standby	1
Off	0

Transition Times	
Operational → Off	0:05
Off → Operational	0:05
Operational → Standby	0:10
Standby → Operational	0:10
Standby → Off	0:10
Off → Standby	0:10

State Transition

State: Operational

3D OK Cancel Apply

- The second machine, *Station1*, processes the parts for 2 minutes each. We entered the kilowatts it uses up in the states it can be in. We also entered the times that the machine requires for switching from one energy state to another.

.Models.MyPlant.Station1

Navigate View Tools Tabs Help

Name: Station1 ☐ Failed ☐ Entrance Locked ☐

Label: ☐ Planned ☐ Exit Locked ☐

Times Set-Up Failures Controls Exit Statistics Importer **Energy** Cost

☒ Active ☐

Power Input [kW]	
Working	2
Setting-up	1
Operational	0.5555
Failed	1
Standby	0.1
Off	0.001

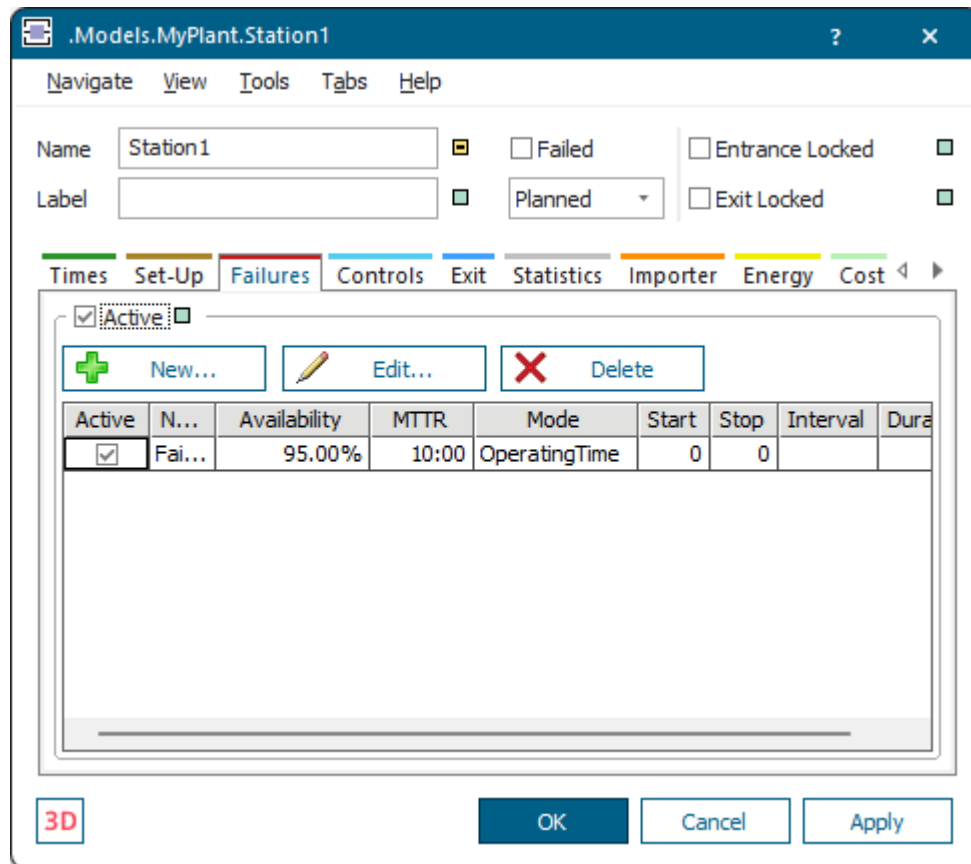
Transition Times	
Operational → Off	1:00
Off → Operational	2:05
Operational → Standby	0:20
Standby → Operational	0:20
Standby → Off	0:20
Off → Standby	0:20

State Transition

State: Operational

3D OK Cancel Apply

To simulate a real situation, we also defined a failure profile with these settings:



- The conveyor, the object of type *Conveyor*, transports the parts from the processing station *Station1* to the *ParallelStation*. We entered the kilowatts it uses up in the states it can be in. We also entered the times that the conveyor requires for switching from one energy state to another. You will notice that the length-oriented object *Conveyor* provides fewer energy settings than the point-oriented objects *Station* and *ParallelStation*.

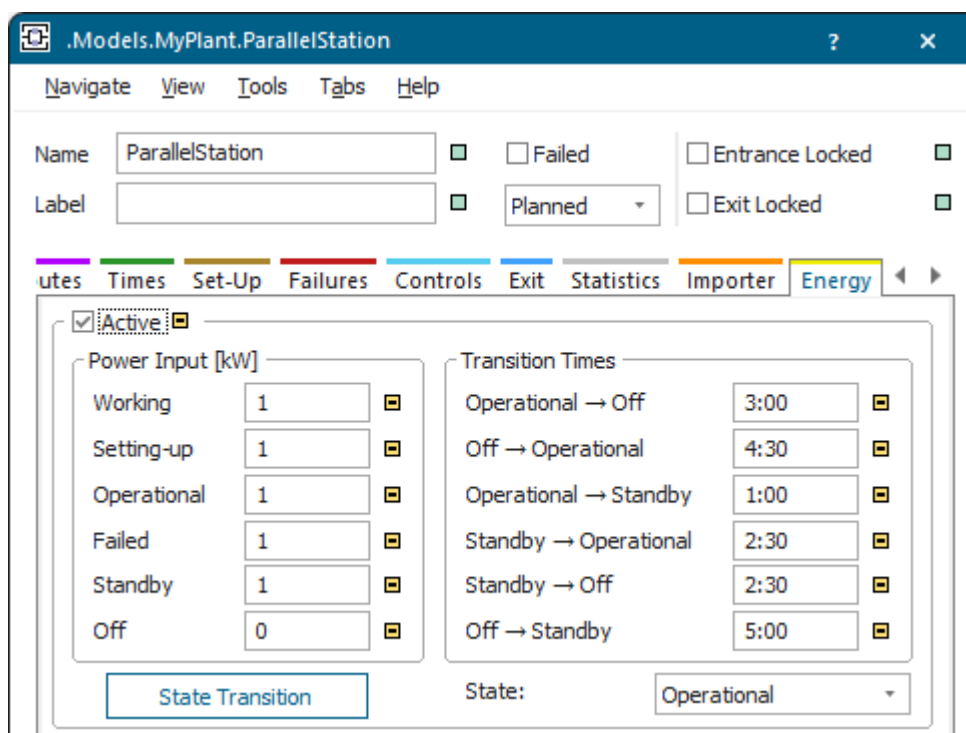
The screenshot shows the configuration window for a conveyor model. The 'Energy' tab is selected. The 'Name' is 'Conveyor' and 'Label' is empty. The 'Planned' dropdown is set to 'Planned'. The 'Active' checkbox is checked. The 'Power Input [kW]' section has the following values: Working (1), Setting-up (0), Operational (0), Failed (1), Standby (0), and Off (0). The 'Transition Times' section has 'Operational → Off' set to 1:00 and 'Off → Operational' set to 1:30. The 'State' dropdown is set to 'Operational'.

To simulate a real situation, we also defined a failure profile with these settings:

The screenshot shows the configuration window for a conveyor model, with the 'Failures' tab selected. The 'Active' checkbox is checked. There are buttons for 'New...', 'Edit...', and 'Delete'. Below these is a table with the following data:

Active	Name	Availability	MTTR	Mode	Start	Stop	Interval
☒	Failure	95.00%	15:00	OperatingTime	0	0	

- The third machine, the *ParallelStation*, processes the parts for 1 minute each. We entered the kilowatts it uses up in the states it can be in. We also entered the times that the machine requires for switching from one energy state to another.



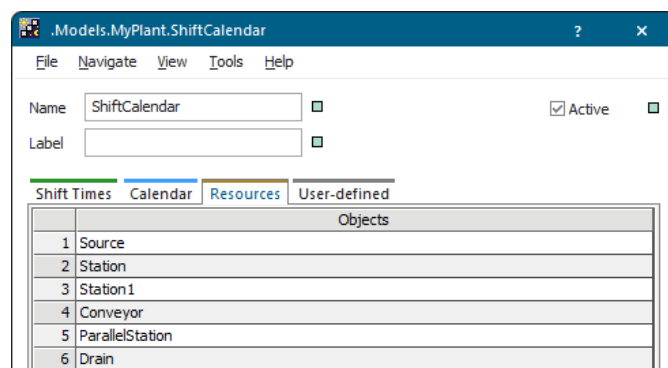
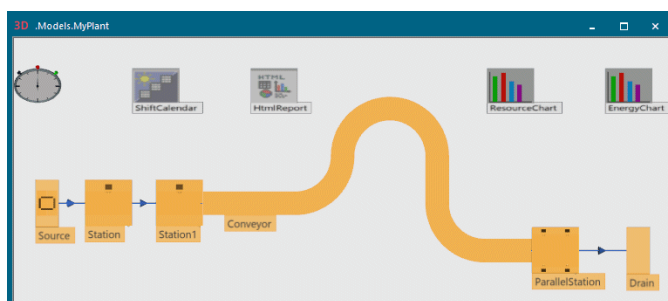
- The *Drain* removes the parts from the factory. We did not change any of its built-in settings.
- The *Charts* display a **resource statistics chart**, and an **energy statistics chart** for the selected objects.

Back to [Simulate the Power Consumption in Your Facility](#)

Configure the ShiftCalendar

To model a realistic situation in the factory, we insert a *ShiftCalendar* that controls the hours during which people and machines work in the factory. We did not change the default shift times, we just entered the material flow objects.

To do so, we held down **Shift** and dragged a marquee over the objects. We then dragged the selected objects onto the *ShiftCalendar* and dropped them there.

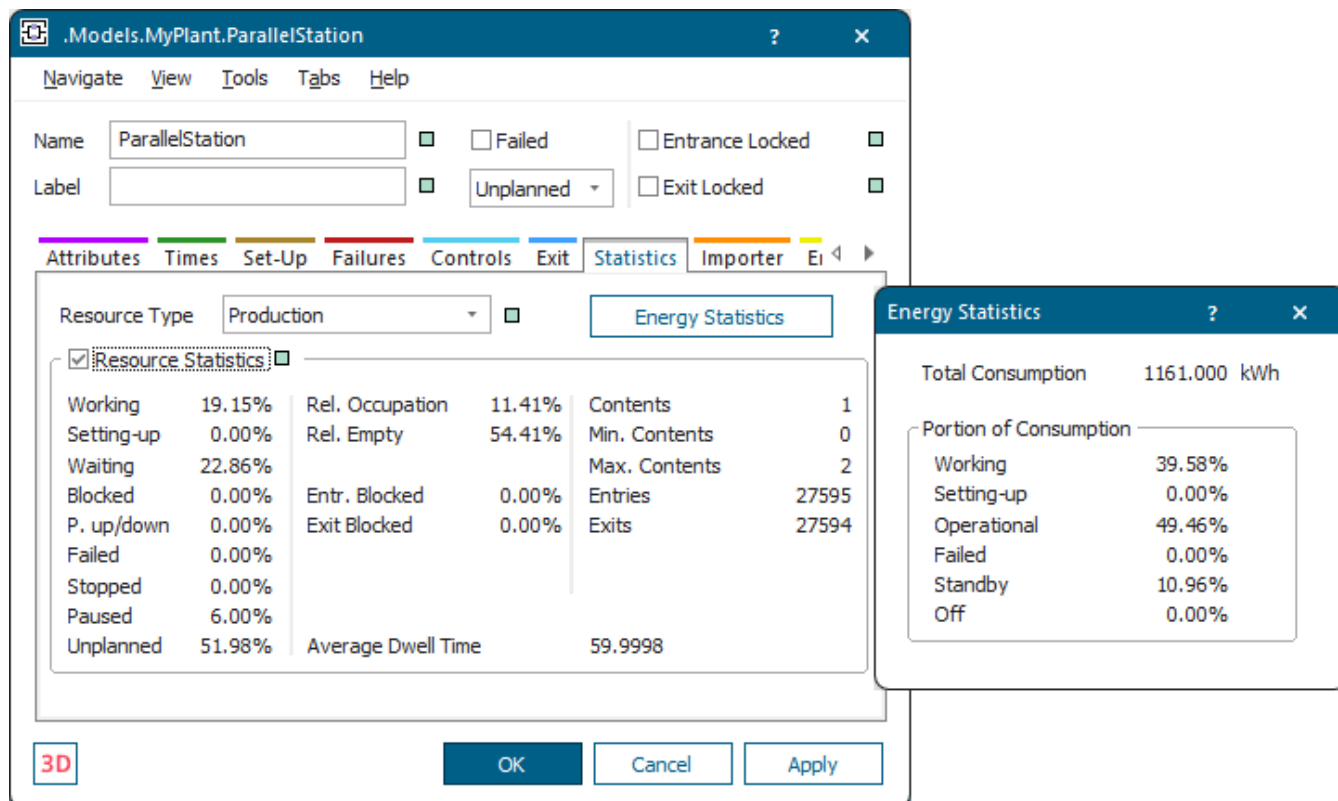


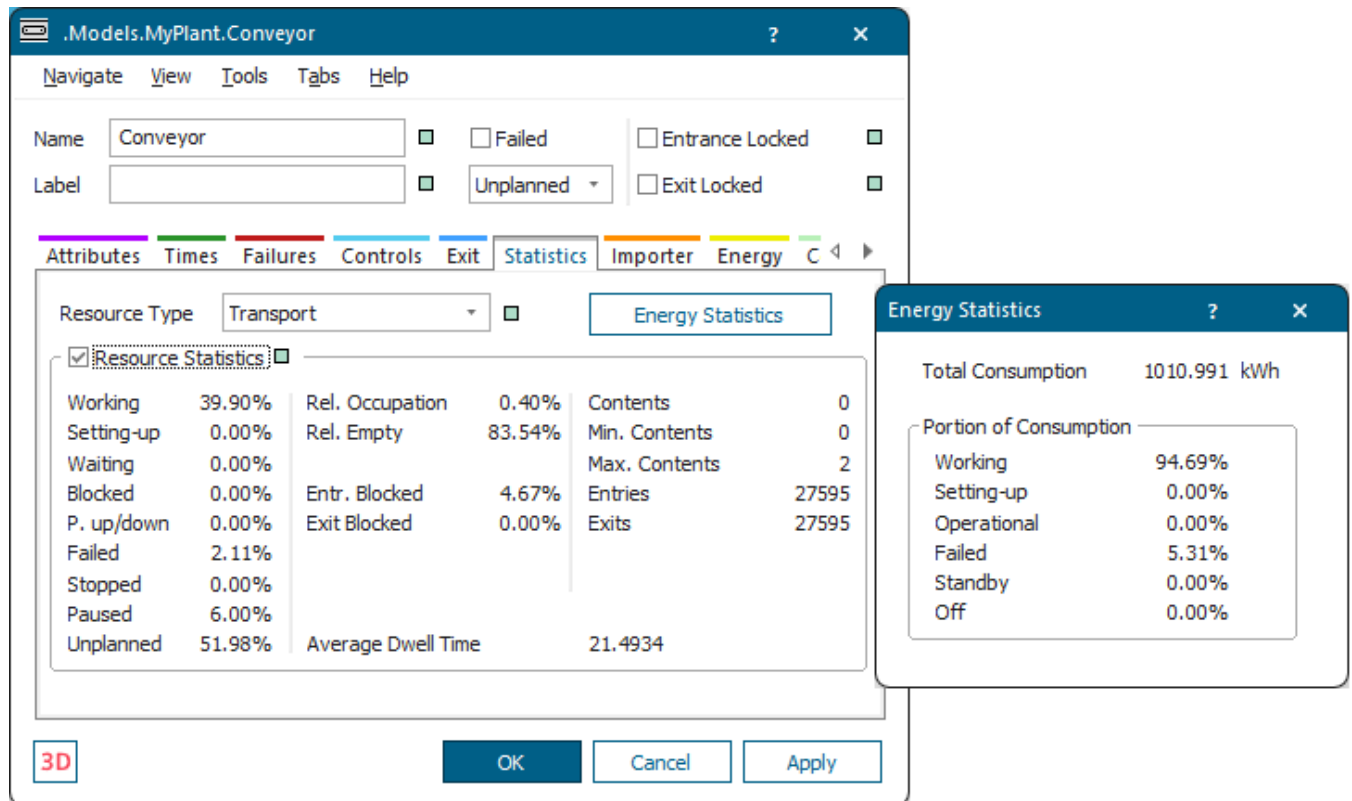
Back to Simulate the Power Consumption in Your Facility

Check the Power Consumption in the Dialogs of the Objects

You can view the energy consumption values, which the objects collected, in their dialogs.

- Double-click the icon of the respective object and click the **Tab Statistics**.
- Click **Energy Statistics** to view more detailed values about the different portions of energy consumption.



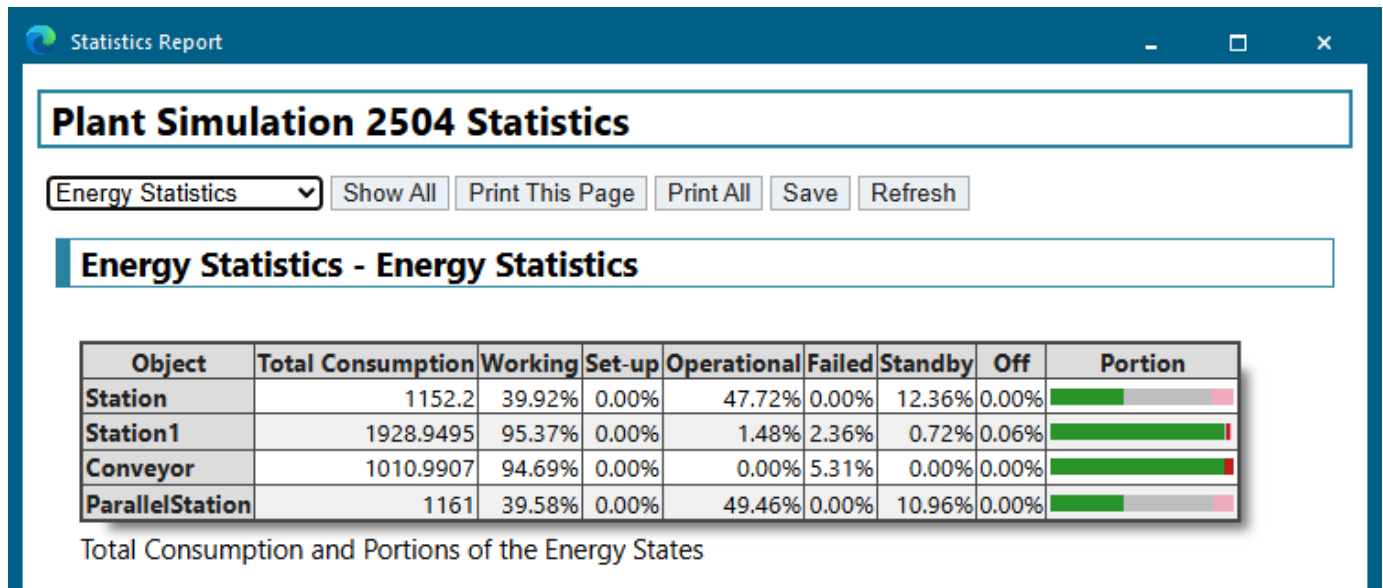


[Back to Simulate the Power Consumption in Your Facility](#)

Check the Power Consumption in the Statistics Report

You can check the power consumption of the material flow objects in your simulation model in the *StatisticsReport*.

To view the **total energy consumption and the portions of the different energy states**, select **Energy Statistics** in the drop-down list of the **Statistics Report**. The values are described under **Energy Statistics — Energy Statistics**.



Note

The energy states of the material flow objects differ from the resource states with the same name. The values for the resource states refer to the statistics collection period, while the values for the energy states refer to the total energy consumption.

The material flow objects also show these values on the [Tab Statistics](#) and in the dialog that clicking [Energy Statistics](#) opens.

[Back to Simulate the Power Consumption in Your Facility](#)

Check the Power Consumption in the Chart

You can view the energy consumption values, which the objects collected, in a *Chart*.

- To view the **energy consumption of the different energy states**, select the objects which you want to display in the *Chart*. Drag them onto the [Chart](#). Select the **Statistics Type > Energy Statistics** in the dialog that opens and click **OK**.